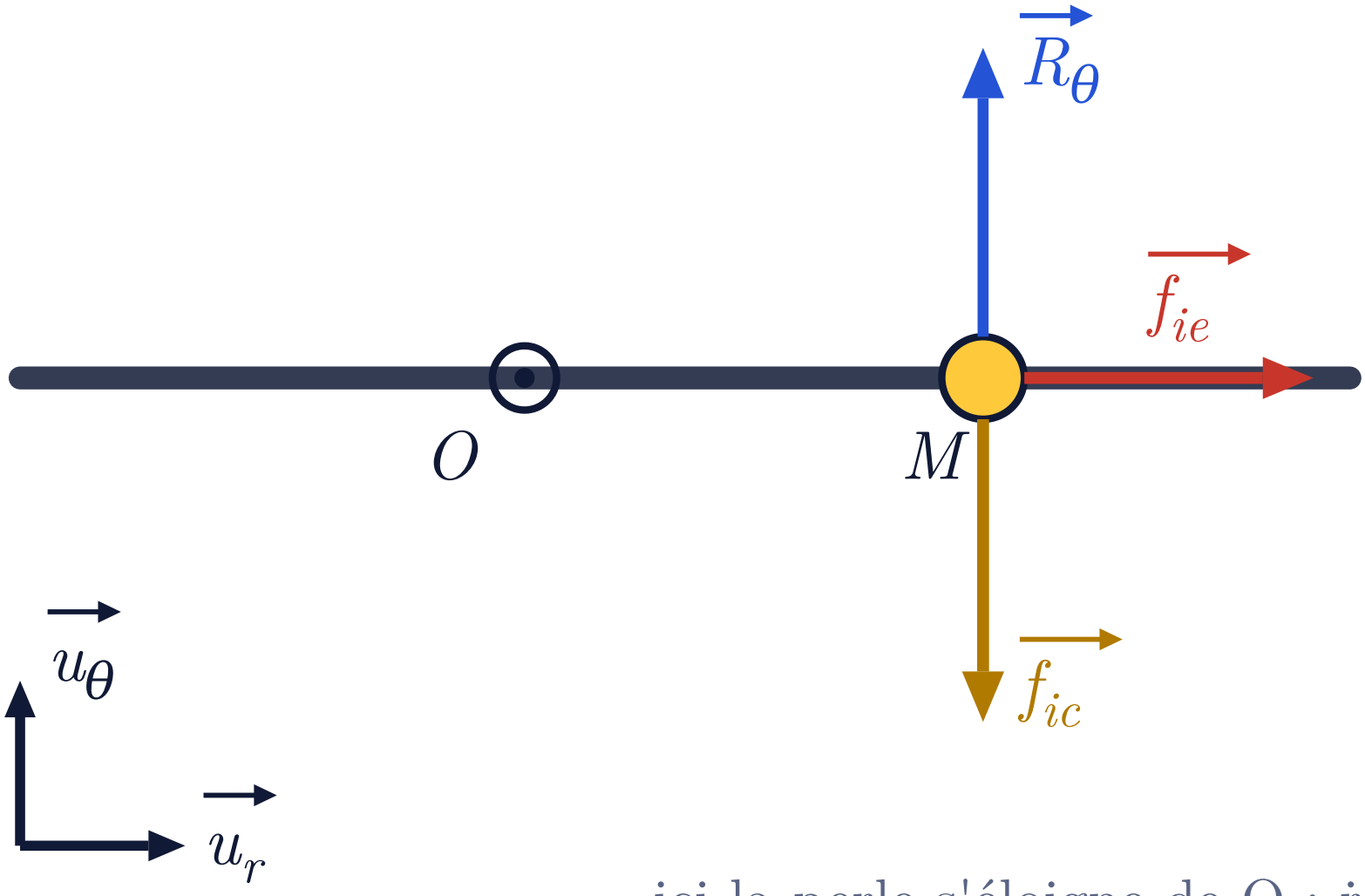

ici la perle s'éloigne de O : $\dot{r} > 0$